

A solution of Erdős Problem 289 with intervals of length two and three

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Abstract

Erdős and Graham asked whether, for every sufficiently large integer k , there exist k finite intervals of positive integers, each containing at least two integers and no two of them overlapping or adjacent, such that the reciprocals of all integers in the intervals sum to 1. We prove a stronger statement: for every sufficiently large k the intervals can be chosen to have length two or three and to lie in $[1, 20k]$. The proof adapts prime-power absorption to signed pairs of consecutive integers. Complementary modular inverses and an elementary sieve provide large correction fibers, and a finite orientation argument makes these fibers simultaneously nonadjacent. A sparse inverse-covering lemma, obtained from the Conlon–Fox–Pham structure theorem and a divisor-count argument, enables a descending denominator correction with a predetermined interval count. A protected core is completed using the Liu–Sawhney theorem. The accompanying Lean formalization proves the stated target with no custom axioms in its dependency chain.

1 Introduction

1.1 The problem and the main theorem

Problem 289 on the Erdős problems site [EP], originating in Erdős and Graham [EG], reads as follows.

Is it true that, for all sufficiently large k , there exist finite intervals $I_1, \dots, I_k \subset \mathbb{N}$, distinct, not overlapping or adjacent, with $|I_i| \geq 2$ for $1 \leq i \leq k$, such that

$$1 = \sum_{i=1}^k \sum_{n \in I_i} \frac{1}{n} ?$$

Throughout, an interval of integers $[a, b]$ with $a \leq b$ consists of all integers n with $a \leq n \leq b$, so $|[a, b]| = b - a + 1$. Two intervals $[a, b]$ and $[a', b']$ with $b < a'$ are *adjacent* if $b + 1 = a'$, and *separated* if $b + 1 < a'$, that is, if at least one integer lies strictly between them. A family of intervals is separated if any two of its members are separated after ordering them.

Theorem 1. *For every sufficiently large integer k there are integer intervals $[a_i, b_i]$, $1 \leq i \leq k$, such that*

$$2 \leq a_i \leq b_i \leq 20k, \quad b_i - a_i + 1 \in \{2, 3\}, \quad b_i + 1 < a_{i+1} \quad (1 \leq i < k), \quad \sum_{i=1}^k \sum_{n=a_i}^{b_i} \frac{1}{n} = 1. \quad (1.1)$$

Corollary 1. *The answer to Problem 289 is affirmative: for all sufficiently large k there exist finite intervals $I_1, \dots, I_k \subset \mathbb{N}$, distinct, not overlapping or adjacent, with $|I_i| \geq 2$, such that $1 = \sum_i \sum_{n \in I_i} 1/n$.*

Proof. Take $I_i = [a_i, b_i]$ from Theorem 1. For $i < j$ the inequalities $b_i + 1 < a_{i+1} \leq a_j$ show that I_i and I_j are disjoint, hence distinct, and that the integer $b_i + 1$ lies in neither of them, so they are not adjacent. Each $|I_i| \in \{2, 3\}$ is at least 2, and the reciprocal sum is the same. $\square$

The separation requirement is the substance of the problem. Allowing adjacency would permit splitting an interval of length at least four into adjacent pieces of lengths two and three, increasing the interval count without changing the union or its reciprocal mass; and as observed in the discussion of [EP], without the restrictions the question is easy. Theorem 1 says more than the problem asks: the intervals have length two or three, and all of them lie in $[1, 20k]$. The constant 20 is not optimized; it arises from the far band $[4k, 20k]$ used in Section 7, where the only requirement is the inequality $16\gamma > 1$ in (7.2).

Remark 1. The condition $a_i \geq 2$ in (1.1) is automatic once the other conditions hold. If some interval began at 1, its length condition would make it contain both 1 and 2, so its reciprocal mass alone would be at least $1 + \frac{1}{2} > 1$, while every other interval has positive mass; this contradicts total mass one. The formal target in Appendix A requires only $a_i \geq 1$, and this observation bridges the two formulations. The construction below also gives $a_i \geq 2$ directly.

1.2 Outline of the proof

The proof adapts the absorption argument in Section 4 of Conlon, Fox, He, Mubayi, Pham, Suk and Verstraëte [CFHMPSV]. Its descending cancellation of denominator prime powers, its small reciprocal budget, and its terminal reservoir form the starting architecture, and Lemma 3 below is a sparse version of that paper’s inverse-covering theorem, proved by the same bounded-rank progression and divisor-count method. The new correction mechanism uses pairs of consecutive integers with complementary signs, a finite orientation selection that makes them simultaneously nonadjacent, a fixed number of pairs at each correction stage, and a core completed by extending pairs to triples.

The final family consists of intervals of three kinds, living at three different scales (Figure 1).

1. **Core pairs** $[Kd+1, Kd+2]$, where K is a fixed integer and d runs over a set $\mathcal{R}$ of parameters of size $\Theta(k^{4/5})$ (Section 6). Some of these pairs will later be extended to the triples $[Kd, Kd+2]$.
2. **Main pairs** $[a, a+1]$ with $3 \mid a$, lying in $[k/1024, k] \cup [4k, 20k]$, all of whose endpoints are $k^{4/5}$ -powersmooth (Section 7). There are exactly $k - |\mathcal{R}| - C_H$ of them, where C_H is a number fixed in advance, and they are chosen so that the *initial deficit*

$$r_0 = 1 - (\text{mass of core}) - (\text{mass of main pairs})$$

is a rational number of size about $\delta/2$, with $\delta = 1/1000$, whose denominator has all prime-power divisors at most $H = O(k^{4/5})$.

3. **Correction and auxiliary pairs**, with endpoints between L and $8H^{11/10} + 1 = O(k^{22/25})$, added by a descent (Section 5) which cancels the prime powers of the denominator of r_0 one at a time, in decreasing order.

The descent works as follows. Let $q = p^a$ be the largest prime power in the current denominator. A *signed correction pair* at q is a pair $\{qm, qm + \sigma\}$ of consecutive integers, with a multiplier m coprime to p and a sign $\sigma \in \{-1, +1\}$, such that the neighbour $qm + \sigma$ has all its prime-power divisors below q . Its mass is

$$w_\sigma(qm) = \frac{1}{qm} + \frac{1}{qm + \sigma},$$

and modulo q the first term behaves like m^{-1}/q while the second is a junk term whose denominator involves only smaller prime powers. Lemma 1 produces, by an elementary construction, a fiber of about q^ε such multipliers: for each auxiliary parameter t the two complementary multipliers m_1 and m_2 with $m_1 + m_2 = t$ (or $4t$) satisfy $qm_1 \pm 1 = rt$ and $qm_2 \mp 1 = (q - r)t$ for a modular inverse r (with t replaced by $4t$ throughout in the odd-prime case), so one of them is coprime to p , and a sieve on t together with the divisor bound makes the neighbour smooth. No equidistribution of modular inverses is needed. Lemma 3 shows that the inverses of a sparse set of size $q^{7\varepsilon/8}$ already cover every residue modulo q using at most $q^{\varepsilon/2}$ summands. Subtracting the corresponding pair masses therefore removes the factor p^a from the denominator while introducing only smaller prime powers. This is the obstacle peculiar to pairs, as opposed to single unit fractions: the junk term $1/(qm + \sigma)$ must not reintroduce large prime powers.

Signed pairs at different labels could overlap or touch. Each pair has one endpoint that is a multiple of four, its *position*, and is either the left pair $[y - 1, y]$ or the right pair $[y, y + 1]$ at that position y . Lemma 2 chooses, for all the finitely many prime powers $q \leq H$ at once, one orientation per position by a finite counting argument, and keeps at least a quarter of every fiber; the retained pairs are then pairwise separated across all labels. This choice is finite and is made after H is known.

To make the total number of intervals predetermined, Lemma 5 reserves, for each prime power q , a family $\mathcal{F}_q$ of exactly $s(q) = \lfloor q^{1/20} \rfloor$ *auxiliary* pairs whose endpoints are $(q/2)$ -powersmooth. If the cancellation at q uses $c \leq s(q)$ correction pairs, $s(q) - c$ auxiliary pairs are added as well. Every stage then adds exactly $s(q)$ intervals, and the whole descent adds exactly $C_H = \sum_{L < q \leq H} s(q)$ intervals. No interval is ever split or merged.

After the descent the residual deficit is j/K for some integer $1 \leq j \leq j_0$, with j_0 fixed. It is absorbed by extending core pairs to triples: extending $[Kd + 1, Kd + 2]$ to $[Kd, Kd + 2]$ adds $1/(Kd)$, and the Liu–Sawhney theorem (through Lemma 6) supplies, inside each of j disjoint bands of parameters, a set D with $\sum_{d \in D} 1/d = 1$. This adds exactly j/K without changing the number of intervals.

Separation is arranged at each scale: retained correction pairs are separated by Lemma 2, auxiliary pairs are chosen at distance at least two from every possible signed correction endpoint (Lemma 5), core parameters d are only used when the neighbourhood $[Kd - 1, Kd + 3]$ avoids every possible correction and auxiliary endpoint, and main pairs lie on a grid of spacing three, far beyond every other endpoint. The auxiliary families and the core are protected against a deterministic superset $\mathcal{P}_L^*$ of all possible signed correction endpoints, which has density zero (Proposition 3), so they do not depend on the finite orientation choice. The final assembly in Section 8 verifies the count, the mass and the conditions in (1.1). Figure 2 shows how the ingredients depend on one another.

An earlier draft of this paper obtained the correction fibers from the Bourgain–Garaev bound for short sums of modular inverses [BG] and the Erdős–Turán inequality. The signed construction of Section 3 removes that input entirely; the structure theorem of Section 2.1 is still used.

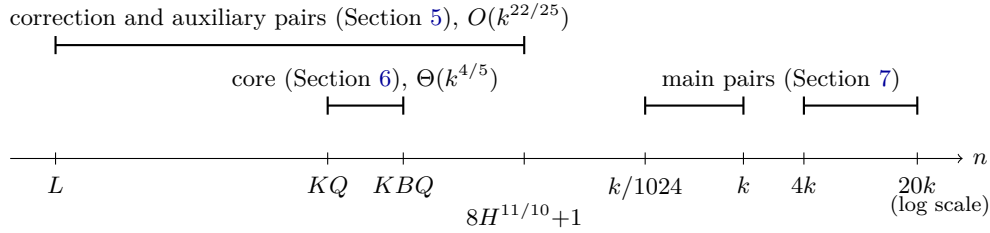


Figure 1: Where the three kinds of intervals live. Here $Q = \lfloor k^{4/5} \rfloor$, $H = KBQ + 2$, and K, B, L are constants.

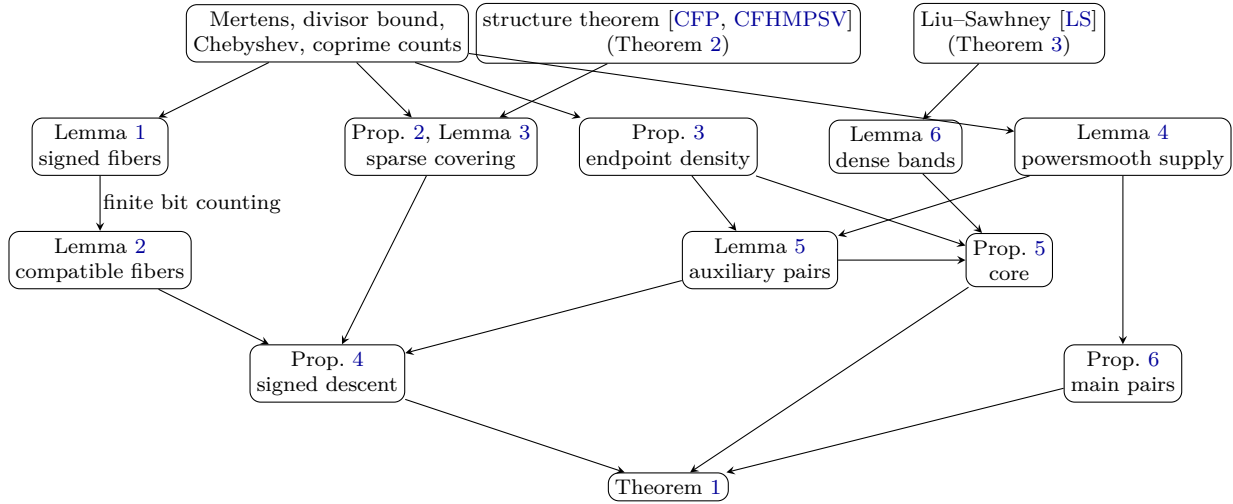


Figure 2: Dependencies among the main statements.

1.3 Notation and conventions

An integer is *y-powersmooth* if every prime-power divisor of it is at most y , and *strictly q-powersmooth* if every prime-power divisor of it is less than q ; for integral q the latter is $(q - 1)$ -powersmoothness. The largest prime-power divisor of an integer always refers to numerical size. For a positive integer a write

$$w(a) = \frac{1}{a} + \frac{1}{a+1},$$

and for a signed correction pair, with $qm \geq 2$ and $\sigma \in \{-1, +1\}$,

$$w_\sigma(qm) = \frac{1}{qm} + \frac{1}{qm + \sigma}, \quad J(q, m, \sigma) = [\min(qm, qm + \sigma), \max(qm, qm + \sigma)].$$

The *mass* of a set of pairwise disjoint intervals is the sum of the reciprocals of all integers in them; thus the pair $[a, a + 1]$ has mass $w(a)$ and $J(q, m, \sigma)$ has mass $w_\sigma(qm)$. All logarithms are natural, except for the explicitly written $\log_2$ in the structure theorem and its application. $\pi(x)$ counts primes up to x , and $\tau(n)$ is the number of divisors of n . A prime power is an integer p^a with p prime and $a \geq 1$; sums and unions over q “prime power” in a range run over all such integers in that range. Asymptotic notation $o(\cdot)$ and $O(\cdot)$ refers to $q \rightarrow \infty$, $X \rightarrow \infty$ or $k \rightarrow \infty$, as indicated, with every constant of Section 2.4 held fixed. The symbol $M_q = q^\varepsilon$ is used for the fiber scale, and $M = k - R - C_H$ for the number of main pairs; the function $R(q)$ of Section 3.1 is distinct from the core count $R = |\mathcal{R}|$.

1.4 Formal verification

The theorem has been formalized in Lean 4 using Mathlib. The certificate is recorded in the accompanying repository [Repo] at commit 61f54d7 of its main branch (full hash in Appendix A.2). The tag v1.0-unconditional-certificate records an earlier certificate for the same terminal target; Appendix A.2 identifies the later checked revision. The terminal declaration `Erdos289.candidateStatement` has type `Erdos289.CandidateStatement`. Its transitive axiom report contains only `propext`, `Classical.choice` and `Quot.sound`, the usual foundational axioms; no custom analytic axiom occurs.

The formal proof uses the signed-fiber construction and the progression-based inverse-covering argument presented in this paper. The structure theorem, the Liu–Sawhney theorem, Mertens’ theorem and the divisor bound are supplied by formal proofs, with bridges to the statements used here; Chebyshev’s bound comes from Mathlib. Bourgain–Garaev and Erdős–Turán are not used. The original module containing six assumed inputs is retained in the repository as an archival specification, but none of its assumed declarations belongs to the terminal theorem’s dependency chain.

The target definition is the one fixed in the statement audit identified in Appendix A. It requires exactly k intervals of lengths two or three, endpoints at most $20k$, an omitted integer between every ordered pair of intervals, and exact rational reciprocal mass one; see Remark 1 for the lower endpoint. The formalization of Problem 289 in the Formal Conjectures library [FC] asks only for disjoint intervals and omits adjacency; Theorem 1 implies both readings. Appendix A identifies the source revision, toolchain, dependency revisions, target-file hash, terminal theorem and axiom report. The kernel checks the formal implications; the correspondence between the formal proposition and the problem as posed is a human check, recorded there.

2 External inputs and choice of parameters

The proof uses two theorems from the literature and three classical estimates. Each is stated here in the form in which it is used, together with the place where its hypotheses are verified.

2.1 Structure in bounded subset sums

A *represented* generalized arithmetic progression of rank D is given by integer generators $d_1, \dots, d_D$ and real bounds $\alpha_i \leq \beta_i$; for $t > 0$ its *dilate at scale t* is

$$tP = \left\{ \sum_{i=1}^D n_i d_i : n_i \in \mathbb{Z}, t\alpha_i \leq n_i \leq t\beta_i \right\},$$

and $P = 1P$. The generators are retained under dilation; only the coordinate intervals expand. For noninteger t this depends on the representation, which is kept fixed throughout. The dilate tP is *proper* if its coordinate box $\prod_i ([t\alpha_i, t\beta_i] \cap \mathbb{Z})$ is nonempty and distinct coordinate vectors in it give distinct integers. Properness here includes nonemptiness: an empty coordinate box would be vacuously injective, and the face argument in Lemma 3 needs an integer point to fix the inactive coordinates.

Theorem 2 (Conlon–Fox–Pham [CFP, Theorem 1.5]; Conlon–Fox–He–Mubayi–Pham–Suk–Verstraëte [CFHMPSV, Theorem 3]). *Given fixed $\beta > 1$ and $0 < \eta < 1$, there are constants $c > 0$, d_0*

and $m_0 \geq 2$ with the following property. If $m \geq m_0$, $A \subseteq [1, n]$ with $|A| = m$, $n \leq m^\beta$, and

$$m^\eta \leq s \leq \frac{cm}{\log_2 m},$$

then there are a represented progression P of rank at most d_0 , a set $J \subseteq A$, and a set $J' \subseteq J$, such that

$$|J| \geq m - \frac{s \log_2 m}{c}, \quad J \cup \{0\} \subseteq P, \quad |J'| \leq s,$$

P and csP are proper (in particular nonempty), and the subset sums of J' contain a translate of csP , with dilation taken in the supplied representation.

The logarithms in Theorem 2 are to base 2, following the convention of [CFP]; the change of base only affects the constant c . Nonemptiness of the dilate is part of what the source proves: the representation supplied in the proof of [CFP, Theorem 1.5, pp. 26–27] is centered, and a centered coordinate box contains the zero vector after every positive dilation. We cite the numbering of the journal version of [CFHMPSV], which agrees with arXiv version 2; the numbering differs in the first arXiv version. Theorem 2 is used only in the proof of Lemma 3.

2.2 Dense sets contain a unit-fraction sum of one

Theorem 3 (Liu–Sawhney [LS, Theorem 1.3]). *For any fixed $0 < \zeta < 1/2$, every sufficiently large N and every $A \subseteq [1, N]$ with*

$$|A| \geq (1 - 1/e + \zeta)N$$

admit $D \subseteq A$ with $\sum_{d \in D} 1/d = 1$.

Theorem 3 is used only in the proof of Lemma 6.

2.3 Elementary estimates

We use Chebyshev's bound $\pi(x) \ll x/\log x$, Mertens' estimate

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + B_1 + o(1) \tag{2.1}$$

for a constant B_1 , and the divisor bound: for every fixed $\xi > 0$, $\tau(n) \leq n^\xi$ for all sufficiently large n ; see [HW, Theorems 7, 427 and 315]. The divisor bound is applied to integers of size up to a constant times $q^{1+\varepsilon}$, and it is convenient to state the uniform form once.

Proposition 1 (divisor envelope). *Fix $\varepsilon > 0$, $c_0 \geq 1$ and $\xi > 0$. For all sufficiently large q ,*

$$\max_{1 \leq n \leq \lfloor c_0 q^{1+\varepsilon} \rfloor} \tau(n) \leq q^\xi.$$

Proof. Put $\xi' = \xi/(2(1+\varepsilon))$ and let N_0 be such that $\tau(n) \leq n^{\xi'}$ for $n \geq N_0$. For $n < N_0$, $\tau(n) \leq n < N_0 \leq q^\xi$ once q is large. For $N_0 \leq n \leq c_0 q^{1+\varepsilon} + 1 \leq 2c_0 q^{1+\varepsilon}$,

$$\tau(n) \leq (2c_0)^{\xi'} q^{(1+\varepsilon)\xi'} = (2c_0)^{\xi'} q^{\xi/2} \leq q^\xi$$

for large q . Different applications below use different values of ξ ; each is a separate instance of this proposition. $\square$

2.4 Choice of parameters

All constants are chosen once, in the following order, and are independent of k .

1. $\varepsilon = 1/10$. Lemmas 1, 2 and 3 and Proposition 3 are proved for every fixed $\varepsilon \in (0, 1)$; only this value is used afterwards.
2. $\delta = 1/1000$, the approximate size of the initial deficit.
3. An initial cutoff L_{base} , large enough that the signed construction of Section 3 is defined for all prime powers $q > L_{\text{base}}$ and Proposition 3 holds for the endpoint set $\mathcal{P}_{L_{\text{base}}}^*$. Since $\mathcal{P}_L^* \subseteq \mathcal{P}_{L_{\text{base}}}^*$ for every $L \geq L_{\text{base}}$, the density statement and the auxiliary-supply threshold derived from it in Lemma 5 apply uniformly to every later cutoff. This avoids choosing a threshold that depends on the cutoff it must bound.
4. An integer $L \geq L_{\text{base}}$, so large that all of the following hold.
 - (a) $L \geq 11$.
 - (b) Lemma 1 applies to every prime power $q > L$, and Lemma 2 applies with cutoff L (that is, $L \geq L_0$ there).
 - (c) Lemma 3 applies with $C = 8$ to every prime power $q > L$.
 - (d) The size conditions of the signed construction hold for $q > L$: $q > 8q^{1/10}$, $R(q) \geq 1$ and $E(q) \geq 1$, with R and E as in Section 3.1.
 - (e) The divisor-envelope thresholds used in the sieve of Lemma 1 hold for $q > L$.
 - (f) $E(q) \leq q^{1/40}$ for every $q > L$.
 - (g) Lemma 5 applies with cutoff L .
 - (h) $120L^{-1/40} < \delta/16$.
 - (i) $K := \text{lcm}(1, \dots, L)$ satisfies $K\delta \geq 2$ and $K \geq 8$.

Each condition holds for all sufficiently large L , so a single choice of L meets all of them; condition (a) alone is not sufficient for the construction.

5. $K = \text{lcm}(1, \dots, L)$, $j_0 = \lceil K\delta \rceil$, and $B = 4^{j_0}$.
6. The deterministic endpoint set $\mathcal{P}_L^*$ of Section 5.2, and for every prime power $q > L$ a family $\mathcal{F}_q$ of $s(q) = \lfloor q^{1/20} \rfloor$ auxiliary pairs from Lemma 5; these determine $\mathcal{F}^*$ and $\mathcal{U} = \mathcal{P}_L^* \cup \mathcal{F}^*$. The individual signed fibers A_q of Lemma 1 may also be fixed now; no globally compatible infinite family of oriented pairs is required or asserted.

The quantities depending on k are then determined in this order:

$$Q = \lfloor k^{4/5} \rfloor, \quad H = KBQ + 2, \quad C_H = \sum_{\substack{L < q \leq H \\ q \text{ prime power}}} s(q),$$

the compatible fibers $(I_q, \sigma_q)_{L < q \leq H}$ of Lemma 2 for this H ,

$$\mathcal{R} \text{ and } R = |\mathcal{R}| \text{ (Section 6),} \quad M = k - R - C_H.$$

The core $\mathcal{R}$ avoids all of $\mathcal{P}_L^*$ and therefore does not depend on the orientation choice; the two may be made in either order. The phrase “for sufficiently large k ” is used at finitely many places, each time with a threshold depending only on the constants above. The argument yields no explicit value of that threshold, and the constants K and B , as well as the thresholds implicit in Theorems 2 and 3, are extremely large.

3 Signed correction fibers

Throughout this section $0 < \varepsilon < 1$ is fixed and $q = p^a$ is a prime power tending to infinity.

3.1 The divisor envelope and the multiplier range

For sufficiently large integers q define

$$M_q = q^\varepsilon, \quad V(q) = \max_{1 \leq n \leq \lceil 25q^{1+\varepsilon} \rceil} \tau(n), \quad E(q) = V(q)^2 \log q, \quad R(q) = \frac{q^\varepsilon}{E(q)}. \quad (3.1)$$

By Proposition 1, $V(q) \leq q^\xi$ for every fixed $\xi > 0$ and large q ; hence

$$V(q) = q^{o(1)}, \quad E(q) = q^{o(1)}, \quad R(q) = q^{\varepsilon - o(1)} \rightarrow \infty. \quad (3.2)$$

The factor $\log q$ in $E(q)$ ensures $E(q) \rightarrow \infty$ without a separate argument that $V(q)$ does. The constant 25 is a harmless envelope constant: every integer to which $V(q)$ is applied below is a neighbour $qm + \sigma$ with $1 \leq m \leq 8M_q$, and

$$qm + \sigma \leq 8qM_q + 1 \leq 9q^{1+\varepsilon} \leq \lceil 25q^{1+\varepsilon} \rceil. \quad (3.3)$$

3.2 Individually valid signed pairs

Lemma 1 (signed fibers). *For every fixed $0 < \varepsilon < 1$ there is q_0 such that every prime power $q = p^a \geq q_0$ admits a finite set A_q of positive multipliers, signs $\sigma_q(m) \in \{-1, +1\}$ and positions $y_q(m) \in 4\mathbb{N}$, $m \in A_q$, satisfying*

$$|A_q| \geq \frac{M_q}{8V(q)}, \quad R(q) \leq m \leq 8M_q, \quad p \nmid m, \quad (3.4)$$

$$qm + \sigma_q(m) \text{ is strictly } q\text{-powersmooth}, \quad (3.5)$$

$$J(q, m, \sigma_q(m)) \in \{[y_q(m) - 1, y_q(m)], [y_q(m), y_q(m) + 1]\}. \quad (3.6)$$

Within each fiber the positions $y_q(m)$, $m \in A_q$, are distinct. The largest prime power dividing either endpoint of $J(q, m, \sigma_q(m))$ is exactly q . Nothing is asserted about separation between pairs of different fibers.

Proof. Initial parameters. Let

$$\mathcal{T}_q = \{t \in [M_q, 2M_q] \cap \mathbb{N} : \gcd(t, 2p) = 1\}.$$

The density of integers coprime to $2p$ is $\frac{1}{2}(1 - 1/p) \geq \frac{1}{3}$ for odd p and $\frac{1}{2}$ for $p = 2$, so $|\mathcal{T}_q| \geq M_q/3 - O(1)$, uniformly in p . Take q so large that $q > 8M_q$; then all multipliers and endpoints constructed below are positive.

Even prime powers. Let $q = 2^a \geq 4$ and $t \in \mathcal{T}_q$, so t is odd. Let $1 \leq r < q$ be the inverse of t modulo q , and put

$$m_1 = \frac{rt - 1}{q}, \quad m_2 = t - m_1.$$

Since $rt \equiv 1 \pmod{q}$ and $rt > 1$, we have $rt \geq q + 1$, so $m_1 \geq 1$; since $r < q$, $m_1 < t$, so $m_2 \geq 1$. Then

$$qm_1 + 1 = rt, \quad qm_2 - 1 = (q - r)t, \quad m_1 + m_2 = t. \quad (3.7)$$

Both multipliers are positive and less than $2M_q$. Since t is odd, exactly one of m_1, m_2 is odd. Choose that one as m , with $\sigma = +1$ if $m = m_1$ and $\sigma = -1$ if $m = m_2$; then $p = 2 \nmid m$. As $4 \mid q$, the endpoint qm is divisible by four, and we set $y = qm$.

Odd prime powers. Let p be odd and $t \in \mathcal{T}_q$, so $\gcd(4t, q) = 1$. Let $1 \leq r < q$ satisfy $4rt \equiv -1 \pmod{q}$, and put

$$m_1 = \frac{4rt + 1}{q}, \quad m_2 = 4t - m_1.$$

Since $4rt + 1 \equiv 0 \pmod{q}$ and $4rt + 1 > 1$, $m_1 \geq 1$; and $m_1 = 4t$ would force $4t(q - r) = 1$, so $m_1 < 4t$ and $m_2 \geq 1$. Then

$$qm_1 - 1 = 4rt, \quad qm_2 + 1 = 4(q - r)t, \quad m_1 + m_2 = 4t. \quad (3.8)$$

Both multipliers are positive and less than $8M_q$. Since $p \nmid 4t$, at least one of them is coprime to p . Choose such an m , with $\sigma = -1$ if $m = m_1$ and $\sigma = +1$ if $m = m_2$; fix a deterministic rule if both are admissible. The neighbour $qm + \sigma$ is $4rt$ or $4(q - r)t$, hence divisible by four, and we set $y = qm + \sigma$.

In both cases no inverse is required to lie in a prescribed subinterval of its residue range; the whole range $1 \leq r < q$ is used. Each $t \in \mathcal{T}_q$ thus yields one signed multiplier (m, σ) ; the neighbour has the form ft or $4ft$ with $1 \leq f < q$, and is at most $8qM_q$.

First sieve: large prime factors of t . Put

$$\eta = \varepsilon/10000, \quad T_q = q^{\varepsilon - \eta}.$$

Discard every t having a prime factor exceeding T_q . Counting multiples of each such prime in $[1, 2M_q]$, their number is at most

$$2M_q \sum_{\substack{T_q < \ell \leq 2M_q \\ \ell \text{ prime}}} \frac{1}{\ell} + \pi(2M_q) = \left(2 \log \frac{\varepsilon}{\varepsilon - \eta} + o(1)\right) M_q < \frac{M_q}{1000} \quad (3.9)$$

by (2.1) and Chebyshev's bound.

Second sieve: large prime powers in the neighbour. Suppose a prime power $u = \ell^b \geq q$ divides the neighbour ft or $4ft$ of a surviving t . If $\ell \nmid 4t$ then $u \mid f$, which is impossible because $f < q \leq u$. Hence $\ell \mid 4t$, so $\ell \leq T_q$ (for large q , $T_q \geq 2$); and $\ell \neq p$, because the neighbour is congruent to $\sigma \not\equiv 0$ modulo p . There are $O_\varepsilon(T_q \log q)$ prime powers u with $\ell \leq T_q$ in the range $q \leq u \leq 8qM_q + 1$. For each such u and each sign, the congruence $qm \equiv -\sigma \pmod{u}$ has at most one solution $m \in [1, 8M_q]$, because $\gcd(q, u) = 1$ and $8M_q < q \leq u$. Each signed multiplier (m, σ) arises from at most $V(q)$ parameters t , because t divides the neighbour $qm + \sigma$, which lies in the range (3.3). The second sieve therefore discards at most

$$O_\varepsilon(T_q V(q) \log q) = o(M_q) \quad (3.10)$$

parameters. For (3.10) it suffices that $V(q) \leq q^{\eta/4}$ and $\log q \leq q^{\eta/4}$, which hold for large q by Proposition 1; then $T_q V(q) \log q \leq q^{\varepsilon - \eta/2}$. This is a much stronger use of the divisor bound than the envelope estimate needed for the mass bound in Section 5.5.

Third sieve: small multipliers. Discard the parameters whose multiplier satisfies $m < R(q)$. Each m has at most $2V(q)$ preimages t , allowing both signs, so this costs at most

$$2(R(q) + 1)V(q) = \frac{2M_q}{V(q) \log q} + 2V(q) = o(M_q). \quad (3.11)$$

Conclusion. By (3.9)–(3.11), at least $M_q/3 - M_q/1000 - o(M_q) \geq M_q/4$ parameters t remain for large q . Since each signed multiplier has at most $2V(q)$ preimages, the surviving parameters produce at least $M_q/(8V(q))$ distinct multipliers m . Select one signed pair for each; this is A_q , and (3.4)–(3.6) hold by construction: the neighbour of a surviving t has no prime-power divisor $u \geq q$, so it is strictly q -powersmooth.

Positions and labels. In the even case distinct multipliers give distinct positions qm . In the odd case, $qm + \sigma = qm' + \sigma'$ with $m \neq m'$ would force $q \mid \sigma' - \sigma \in \{\pm 2\}$, impossible for odd $q > 2$. Since $m \leq 8M_q < q$ and $p \nmid m$, the largest prime power dividing qm is exactly q ; the neighbour is strictly q -powersmooth. Hence the label q can be recovered from the unordered pair. $\square$

3.3 Finite simultaneous orientation

Lemma 2 (compatible fibers). *For every fixed $0 < \varepsilon < 1$ there is an integer $L_0 \geq 2$ with the following property. For every integer $L \geq L_0$ and every integer $H \geq L$ there are sets and signs*

$$(I_q, \sigma_q)_{L < q \leq H, \text{ } q \text{ prime power}}, \quad I_q \subseteq A_q, \quad |I_q| \geq q^{7\varepsilon/8},$$

with σ_q the signs of Lemma 1, such that the retained intervals $J(q, m, \sigma_q(m))$, over all $m \in I_q$ and all prime powers $L < q \leq H$, are pairwise distinct and pairwise separated.

The statement is finite: it asserts a compatible family for the labels up to H , chosen after H is known, and not a single compatible infinite family. The finite statement is exactly what the descent needs.

Proof. Let $N_q = |A_q|$. By (3.4) and (3.2), choose L_0 so large that for every prime power $q > L_0$,

$$N_q \geq q^{15\varepsilon/16}, \quad N_q/4 \geq q^{7\varepsilon/8}, \quad (9/10)q^{15\varepsilon/16} \leq \frac{1}{2q^2}. \quad (3.12)$$

Fix $L \geq L_0$ and $H \geq L$. If no prime power lies in $(L, H]$ there is nothing to prove, so assume there is one. Consider the finite set S of all positions $y_q(m)$, $m \in A_q$, $L < q \leq H$. To each $y \in S$ assign a bit, choosing either the left interval $[y - 1, y]$ or the right interval $[y, y + 1]$; there are $2^{|S|}$ assignments, and we average over all of them with equal weight. Retain a candidate (q, m) when its interval $J(q, m, \sigma_q(m))$ agrees with the bit at its position.

Fix a label q . Its N_q positions are distinct, so the retained count B_q is a sum of N_q independent bits, each retained with probability $\frac{1}{2}$. Averaging 2^{-B_q} over the assignments and expanding the product gives

$$\mathbb{E}[2^{-B_q}] = \left(\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2}\right)^{N_q} = (3/4)^{N_q},$$

and therefore, since $2^{-B_q} > 2^{-N_q/4}$ whenever $B_q < N_q/4$,

$$\Pr(B_q < N_q/4) \leq 2^{N_q/4} (3/4)^{N_q} < (9/10)^{N_q} \leq \frac{1}{2q^2}, \quad (3.13)$$

using $2^{1/4} \cdot 3/4 < 9/10$ and (3.12). Here $\Pr$ and $\mathbb{E}$ are normalized counts over the finite cube of bit assignments. A union bound over the labels gives total failure weight at most

$$\frac{1}{2} \sum_{q=L+1}^H \frac{1}{q^2} \leq \frac{1}{2} \sum_{q=L+1}^H \frac{1}{q(q-1)} \leq \frac{1}{2L} < 1. \quad (3.14)$$

Hence some assignment retains at least $N_q/4 \geq q^{7\epsilon/8}$ candidates in every fiber; let I_q be the retained multipliers.

Separation. Consider two retained intervals at positions $y < z$. Both are multiples of four, so $z \geq y + 4$. The earlier interval is contained in $[y - 1, y + 1]$ and ends at $b \leq y + 1$; the later starts at $a' \geq z - 1 \geq y + 3$. Thus $b + 1 \leq y + 2 < a'$, the required omitted integer. At a common position only one orientation is retained, and a retained interval at that position and with that orientation is a single interval; two different labels cannot supply the same interval because the largest prime power dividing its endpoints identifies the label (Lemma 1), and within one fiber positions are distinct. So retained intervals at a common position coincide, and distinct retained intervals have distinct positions and are separated. $\square$

The bit selects left or right relative to the position y , not a uniform sign: for $p = 2$ the position is $y = qm$ and $\sigma = +1$ means the right interval, while for odd p the position is $y = qm + \sigma$ and $\sigma = +1$ means the left interval $[y - 1, y]$. Retained left endpoints may differ by as little as three; separation, not a gap of four, is the conclusion.

4 Sparse modular inverse subsets cover all residues

4.1 A pigeonhole lemma

Proposition 2 (simultaneous approximation). *Let $q \geq 2$ and $d \geq 1$, let $d_1, \dots, d_d$ be integers, and let $a_1, \dots, a_d \geq 1$ be real numbers with $V_{\text{GAP}} = \prod_i a_i < q$. Put*

$$B_i = \frac{4q}{a_i} \left(\frac{V_{\text{GAP}}}{q} \right)^{1/d}.$$

Then there are $1 \leq T < q$ and integers $e_i \equiv Td_i \pmod{q}$ with $|e_i| \leq B_i$ for all i .

Proof. Consider the q vectors $(td_i \bmod q)_{1 \leq i \leq d}$, $0 \leq t < q$, in $[0, q)^d$. Ignore the coordinates with $B_i \geq q$, and partition each remaining coordinate range $[0, q)$ into intervals of length B_i , at most $2q/B_i$ of them. Note that $\prod_i (q/B_i) = q/4^d$, and that $B_i \leq 4q$ for every i because $V_{\text{GAP}} < q$ and $a_i \geq 1$. If $r \geq 1$ coordinates remain, the number of boxes is at most

$$\prod_{i: B_i < q} \frac{2q}{B_i} \leq 2^r \cdot \frac{q}{4^d} \cdot 4^{d-r} = q/2^r < q.$$

If no coordinate remains there is one box. In either case two of the q vectors share a box. Their difference, taken with positive sign, has the form $(Td_i \bmod q)_i$ with $1 \leq T < q$, and each coordinate is congruent to an integer of absolute value at most B_i ; ignored coordinates impose no restriction because $B_i \geq q$ there. $\square$

The corresponding step in [CFHMPSV] is formulated with a coprimality hypothesis on the generators d_i . That hypothesis is not available in the setting of Lemma 3, where only the elements of J , not the generators of P , are known to be units modulo q ; the pigeonhole argument above does not need it.

4.2 The covering lemma

Lemma 3 (sparse inverse covering). *For every fixed $0 < \varepsilon < 1$ and $C \geq 1$ there is $q_0(\varepsilon, C)$ such that, for every prime power $q \geq q_0$ and every finite set*

$$I \subseteq [1, Cq^\varepsilon] \cap \mathbb{N}, \quad \gcd(i, q) = 1 \quad (i \in I), \quad |I| \geq q^{7\varepsilon/8},$$

every residue modulo q is a sum of inverses of at most

$$s = \lfloor q^{\varepsilon/2} \rfloor$$

distinct members of I .

This is a sparse extension of [CFHMPSV, Theorem 2]. The proof follows their progression and divisor-count argument, with the density loss accounted for explicitly. The descent uses the lemma with $C = 8$; the threshold q_0 depends on C .

Proof. Structure theorem. Take q so large that $Cq^\varepsilon < q$. Then reduction modulo q , and hence inversion, is injective on I . Represent the inverses of the members of I by integers in $[1, q-1]$; let A be the set of these representatives, of size $m = |I|$, so that

$$q^{7\varepsilon/8} \leq m \leq Cq^\varepsilon.$$

Apply Theorem 2 with $\beta = 2/\varepsilon$, $\eta = 1/3$ and ambient parameter $n = q$. For sufficiently large q ,

$$m \geq m_0, \quad q \leq m^{2/\varepsilon}, \quad m^{1/3} \leq s \leq \frac{cm}{\log_2 m},$$

so we obtain P , J and J' . The structural loss $s \log_2 m/c$ is $o(m)$, so for large q

$$|J| \geq \frac{3}{4}m. \tag{4.1}$$

Keep the representation defining the theorem's dilate fixed:

$$P = \left\{ \sum_{i=1}^D n_i d_i : \alpha_i \leq n_i \leq \beta_i, n_i \in \mathbb{Z} \right\}, \quad D \leq d_0.$$

Choose an integer coordinate vector v representing $0 \in P$. Put $\ell_i = \lceil \alpha_i \rceil$ and $u_i = \lfloor \beta_i \rfloor$. Call a coordinate *active* when $u_i > \ell_i$; all other coordinates are constant on P . Relabel the active coordinates $1, \dots, d$ and put

$$a_i = u_i - \ell_i \geq 1, \quad V_{\text{GAP}} = \prod_{i=1}^d a_i.$$

Every $j \in J$ satisfies

$$j = \sum_{i=1}^d (n_i - v_i) d_i, \quad |n_i - v_i| \leq a_i, \tag{4.2}$$

since the contributions of the constant coordinates vanish after subtracting the representation of zero. There is at least one active coordinate, because P contains zero and the nonzero elements of J .

Upper bound for V_{GAP} . The dilate csP is nonempty by Theorem 2. Fix each inactive coordinate at an integer of its dilated interval, which exists by nonemptiness. This produces a face of that same

proper dilate; no representation is changed and no new dilation is formed. For an active coordinate, its dilated interval contains at least

$$cs(\beta_i - \alpha_i) - 1 \geq csa_i - 1 \geq (cs/2)a_i$$

integers for large q . Thus the face has at least $(cs/2)^d V_{\text{GAP}}$ points. Its supplied translate lies in the subset sums of J' , which lie in $[0, sq]$. Consequently

$$(cs/2)^d V_{\text{GAP}} \leq sq + 1, \quad V_{\text{GAP}} \ll_{\varepsilon} qs^{1-d} \ll_{\varepsilon} q^{1-(d-1)\varepsilon/2}. \quad (4.3)$$

If $V_{\text{GAP}} \geq q$ this already forces $d = 1$, and the final paragraph below applies. Assume $V_{\text{GAP}} < q$.

Lower bound for V_{GAP} . Apply Proposition 2 to the active generators and the numbers a_i , obtaining $1 \leq T < q$ and $e_i \equiv Td_i \pmod{q}$ with $|e_i| \leq B_i$. Put $x = (V_{\text{GAP}}/q)^{1/d}$. For each $j \in J$, by (4.2) the centered representative h of $Tj \pmod{q}$ satisfies

$$|h| \leq \min\left\{\frac{q}{2}, \sum_i a_i B_i\right\} = \min\{q/2, 4dqx\}.$$

Pair this h with the $i \in I$ for which $j \equiv i^{-1} \pmod{q}$; distinct j give distinct i , so there are at least $|J|$ distinct ordered pairs (i, h) , whether or not multiplication by T is injective on J . They satisfy

$$ih = qz + T, \quad z \in \mathbb{Z}, \quad (4.4)$$

and the product is nonzero because $0 < T < q$. Since $i \leq Cq^{\varepsilon}$,

$$|ih| \leq \frac{C}{2}q^{1+\varepsilon}, \quad |ih| \leq 4Cdq^{1+\varepsilon}x.$$

The second bound and (4.4) confine z to a real interval of length $8Cdq^{\varepsilon}x$, so z has at most $1 + 8Cdq^{\varepsilon}x$ integer possibilities. Define the divisor envelope

$$\Delta_C(q) = \max_{1 \leq n \leq \lceil Cq^{1+\varepsilon} \rceil} \tau(n) = q^{o(1)}$$

(Proposition 1), and take q large enough that $\Delta_C(q) \leq m/4$. For each z , the positive integer i divides the nonzero integer $|qz + T| \leq \lceil Cq^{1+\varepsilon} \rceil$ and determines h , so there are at most $\Delta_C(q)$ ordered pairs (i, h) with that z . Therefore

$$\frac{3}{4}m \leq |J| \leq \Delta_C(q)(1 + 8Cdq^{\varepsilon}x) \leq \frac{1}{4}m + 8Cd\Delta_C(q)q^{\varepsilon}x,$$

so $x \geq m/(16Cd\Delta_C(q)q^{\varepsilon})$ and, using $m \geq q^{7\varepsilon/8}$,

$$V_{\text{GAP}} = qx^d \geq \frac{q m^d}{(16Cd)^d \Delta_C(q)^d q^{d\varepsilon}} \geq \frac{q^{1-d\varepsilon/8}}{(16Cd)^d \Delta_C(q)^d}. \quad (4.5)$$

Because C is fixed and $d \leq d_0$ is bounded, this gives

$$V_{\text{GAP}} \geq q^{1-d\varepsilon/8-o(1)}. \quad (4.6)$$

For $d \geq 2$ this contradicts (4.3), because

$$\frac{(d-1)\varepsilon}{2} - \frac{d\varepsilon}{8} = \frac{(3d-4)\varepsilon}{8} > 0.$$

The rank is bounded independently of q , so these comparisons are uniform. Thus $d = 1$.

Conclusion. By (4.2) every $j \in J$ is a multiple of d_1 . Since these j are units modulo q , $\gcd(d_1, q) = 1$. The translated one-dimensional face of the dilate contained in the subset sums of J' has at least $(c/2)sV_{\text{GAP}}$ terms. If $V_{\text{GAP}} \geq q$ this exceeds q for large q ; otherwise (4.6) gives

$$(c/2)sV_{\text{GAP}} \geq (c/2)q^{1+3\varepsilon/8-o(1)} > q.$$

An arithmetic progression with common difference coprime to q and more than q terms covers every residue modulo q . Every such term is a subset sum of J' , which has at most s members, that is, a sum of at most s inverses of distinct members of I . $\square$

5 The correction and descent mechanism

From now on $\varepsilon = 1/10$, and L, K, δ are as in Section 2.4.

5.1 Powersmooth supply

Lemma 4 (powersmooth supply). *Let $0 < \theta < 1$ and $0 < a < b$ be fixed, let $x \rightarrow \infty$, and let $y = x^{\theta+o(1)}$. The number of integers in $[ax, bx]$ that are not y -powersmooth is at most*

$$((b-a)\log(1/\theta) + o(1))x. \quad (5.1)$$

Proof. The number of integers in $[ax, bx]$ divisible by some prime $p > y$ is at most

$$(b-a)x \sum_{y < p \leq bx} \frac{1}{p} + O(\pi(bx)) = ((b-a)\log(1/\theta) + o(1))x$$

by (2.1) and $\pi(bx) = o(x)$. For each prime $p \leq y$ let u_p be the smallest power of p exceeding y . The number of integers at most bx divisible by some u_p is at most

$$\sum_{p \leq y} \left\lfloor \frac{bx}{u_p} \right\rfloor \leq bx \frac{\pi(y)}{y} = o(x).$$

Every integer that is not y -powersmooth is counted by one of the two bounds. $\square$

5.2 Possible correction endpoints

Define the deterministic set of *possible signed correction endpoints*

$$\mathcal{P}_L^* = \bigcup_{\substack{q > L \\ q \text{ prime power}}} \bigcup_{\substack{m \in \mathbb{N} \\ R(q) \leq m \leq 8q^\varepsilon}} \{qm - 1, qm, qm + 1\}. \quad (5.2)$$

It contains both endpoints of every possible signed correction pair $J(q, m, \sigma)$ with $m \in A_q$, before any orientation is chosen, and it depends neither on H , nor on the bit assignment of Lemma 2, nor on the subsets used in cancellation.

Proposition 3 (density zero). *For every fixed $0 < \varepsilon < 1$ and every cutoff L ,*

$$|\mathcal{P}_L^* \cap [1, X]| = o(X) \quad (X \rightarrow \infty). \quad (5.3)$$

The stronger rate $O(X/\log X)$, which held for the narrower endpoint set of the earlier construction, is not claimed for this enlarged set; every later use needs only density zero.

Proof. Fix $0 < \zeta < \varepsilon$. By (3.2) there is q_1 with $E(q) \leq q^\zeta$ for $q \geq q_1$; the finitely many labels $q < q_1$ contribute only finitely many endpoints. For a label $q \geq q_1$, every endpoint is at least $qm - 1 \geq qR(q) - 1 \geq q^{1+\varepsilon-\zeta} - 1$, so an endpoint at most X forces

$$q^{1+\varepsilon-\zeta} \leq qm \leq X + 1, \quad \text{hence} \quad q \leq Z := (X + 1)^{1/(1+\varepsilon-\zeta)}.$$

Put $Y = X^{1/(1+\varepsilon)}$. Labels $q \leq Y$ contribute at most

$$3 \sum_{\substack{q \leq Y \\ q \text{ prime power}}} 8q^\varepsilon = O\left(\frac{Y^{1+\varepsilon}}{\log Y}\right) = O\left(\frac{X}{\log X}\right)$$

endpoints, by Chebyshev's bound. For $Y < q \leq Z$ the multipliers with $qm \leq X + 1$ number at most $(X + 1)/q + 1$, so these labels contribute at most

$$3(X + 1) \sum_{\substack{Y < q \leq Z \\ q \text{ prime power}}} \frac{1}{q} + 3Z.$$

For prime labels, (2.1) gives

$$\sum_{Y < p \leq Z} \frac{1}{p} = \log \frac{\log Z}{\log Y} + o(1) = \log \frac{1 + \varepsilon}{1 + \varepsilon - \zeta} + o(1).$$

Higher prime powers contribute $o(1)$: indeed

$$\sum_{\substack{p^a > Y \\ a \geq 2}} p^{-a} = O(Y^{-1/2}),$$

since for $p \leq \sqrt{Y}$ the geometric tail of the powers exceeding Y is at most $2/Y$, and for $p > \sqrt{Y}$ the tails are bounded by $2 \sum_{n > \sqrt{Y}} n^{-2}$. Finally $Z = o(X)$ because $\zeta < \varepsilon$. Consequently

$$\limsup_{X \rightarrow \infty} \frac{|\mathcal{P}_L^* \cap [1, X]|}{X} \leq 3 \log \frac{1 + \varepsilon}{1 + \varepsilon - \zeta}.$$

The order of limits is: fix ζ , let $X \rightarrow \infty$, then let $\zeta \downarrow 0$. Equivalently, given $\kappa > 0$, first choose ζ so that the displayed constant is less than $\kappa/2$, then choose X large. This proves (5.3). $\square$

Since $\mathcal{P}_L^* \subseteq \mathcal{P}_{L_{\text{base}}}^*$ for $L \geq L_{\text{base}}$, the density statement for the fixed set $\mathcal{P}_{L_{\text{base}}}^*$ bounds every later $\mathcal{P}_L^*$ uniformly, as used in Section 2.4.

5.3 Auxiliary pairs

Lemma 5 (auxiliary pairs). *For every sufficiently large L one can fix, simultaneously for each prime power $q > L$, a family $\mathcal{F}_q$ of exactly*

$$s(q) = \lfloor q^{1/20} \rfloor$$

pairs $[a, a + 1]$ satisfying

$$q^{11/10} \leq a, \quad a + 1 \leq 2q^{11/10}, \quad 4 \mid a,$$

with both endpoints $(q/2)$ -powersmooth, such that all auxiliary pairs, over all $q > L$, are pairwise separated, and every auxiliary endpoint has distance at least two from every element of $\mathcal{P}_L^*$. In particular every auxiliary pair is separated from every possible signed correction pair. If $\mathcal{F}^*$ is the set of all auxiliary endpoints, then

$$|\mathcal{F}^* \cap [1, X]| = O(X^{21/22}). \quad (5.4)$$

Proof. Construct the families in increasing order of the numerical prime power q . Put $x = q^{11/10}$ and $y = q/2$, so $\theta = 10/11$ in Lemma 4. The candidate pairs $[a, a + 1]$ with $4 \mid a$ contained in $[x, 2x]$ number $x/4 + O(1)$; they are pairwise separated. Each integer that is not y -powersmooth spoils at most one candidate, so at least

$$\left(\frac{1}{4} - \log \frac{11}{10} + o(1)\right)x \quad (5.5)$$

candidates have both endpoints y -powersmooth; the constant is positive.

Delete every candidate having an endpoint within distance one of $\mathcal{P}_L^*$. Each element of $\mathcal{P}_L^* \cap [x - 2, 2x + 2]$ spoils at most four candidates, so by Proposition 3 this deletes only $o(x)$ candidates. Delete also every candidate having an endpoint within distance one of a previously chosen auxiliary endpoint. The number deleted in this second step is at most a constant times

$$\sum_{\substack{q' < q \\ q' \text{ prime power}}} s(q') \leq q^{21/20} = o(q^{11/10}) = o(x).$$

Thus a positive constant times x candidates remain, more than $s(q)$ for all sufficiently large q . Select any $s(q)$ of them as $\mathcal{F}_q$. All estimates are uniform in q , so a single sufficiently large L suffices for the entire ascending construction. Separation between auxiliary pairs of the same q holds because all candidates are separated; between different q it holds by the second deletion. Every auxiliary endpoint has distance at least two from $\mathcal{P}_L^*$ by the first deletion, and a possible signed correction pair has both endpoints in $\mathcal{P}_L^*$, so it is separated from every auxiliary pair.

For (5.4), an auxiliary endpoint at most X has label $q \leq X^{10/11}$, so

$$|\mathcal{F}^* \cap [1, X]| \leq 2 \sum_{\substack{q \leq X^{10/11} \\ q \text{ prime power}}} q^{1/20} = O(X^{21/22}). \quad \square$$

The families $\mathcal{F}_q$ depend only on L ; they are fixed once and for all, independently of k and of every later finite choice of compatible fibers. Put $\mathcal{U} = \mathcal{P}_L^* \cup \mathcal{F}^*$. By (5.3) and (5.4),

$$|\mathcal{U} \cap [1, X]| = o(X). \quad (5.6)$$

5.4 The signed cancellation identity

Remark 2. For a signed correction pair $J(q, m, \sigma)$ with $m \in A_q$, in the ring of rationals whose denominators are coprime to p ,

$$q w_\sigma(qm) = \frac{1}{m} + \frac{q}{qm + \sigma} \equiv m^{-1} \pmod{q}, \quad (5.7)$$

because $p \nmid m$ and $qm + \sigma \equiv \sigma \not\equiv 0 \pmod{p}$, so both denominators are coprime to p . Strict q -powersmoothness of $qm + \sigma$ is a separate matter: it guarantees that the denominator of the second term introduces only prime powers below q . This is the identity behind the descent: subtracting pair masses whose m^{-1} sum to the right residue removes the factor q from the denominator. The proof of Proposition 4 makes this explicit with a common denominator.

5.5 The descent

Proposition 4 (signed descent). *Let L, K, δ and the families $\mathcal{F}_q$ be as in Section 2.4, let $H > L$ be an integer, and let $(I_q, \sigma_q)_{L < q \leq H}$ be compatible fibers for this H from Lemma 2. Let r_0 be a rational number with $r_0 \geq \delta/4$ whose reduced denominator has no prime-power divisor exceeding H . Then there is a set $\mathcal{C}$ of exactly*

$$C_H = \sum_{\substack{L < q \leq H \\ q \text{ prime power}}} s(q) \leq H^{21/20} \quad (5.8)$$

pairs, each either a retained signed correction pair $J(q, m, \sigma_q(m))$ with $m \in I_q$ or a member of $\mathcal{F}_q$, for some prime power $q \in (L, H]$, such that:

- (i) *the pairs in $\mathcal{C}$ are pairwise separated;*
- (ii) *every endpoint of a pair in $\mathcal{C}$ lies in $\mathcal{U}$, is at least L , and is at most $8H^{11/10} + 1$;*
- (iii) *the mass $W_{\mathcal{C}}$ of $\mathcal{C}$ satisfies $W_{\mathcal{C}} \leq 120L^{-1/40} < \delta/16$;*
- (iv) *$r = r_0 - W_{\mathcal{C}}$ satisfies $r \geq r_0 - \delta/16 > 0$, and its reduced denominator divides K .*

Proof. Visit every prime power $q \in (L, H]$ in decreasing numerical order, maintaining a current deficit, initially r_0 . The invariant before stage q is that no prime power larger than q divides the current reduced denominator; it holds initially by hypothesis.

Stage $q = p^a$. If q is the full power of p in the current denominator, write the current deficit as $u/(qv)$ with $p \nmid uv$. The retained fiber I_q satisfies the hypotheses of Lemma 3 with $C = 8$: its members are coprime to q and at most $8q^\varepsilon$, and $|I_q| \geq q^{7\varepsilon/8}$. Lemma 3 therefore selects $S \subseteq I_q$ with $c_q = |S| \leq s(q)$ such that

$$\sum_{m \in S} m^{-1} \equiv uv^{-1} \pmod{q}.$$

Write $\sigma_m = \sigma_q(m)$ and subtract the masses $w_{\sigma_m}(qm)$, $m \in S$, from the deficit. To see the effect, let

$$D = \text{lcm}(v, \{m : m \in S\}, \{qm + \sigma_m : m \in S\}).$$

All these integers are coprime to p : $p \nmid v$, $p \nmid m$, and $qm + \sigma_m \equiv \sigma_m \not\equiv 0 \pmod{p}$. The exact calculation is

$$qD \left(\frac{u}{qv} - \sum_{m \in S} w_{\sigma_m}(qm) \right) = \frac{uD}{v} - \sum_{m \in S} \frac{D}{m} - q \sum_{m \in S} \frac{D}{qm + \sigma_m}, \quad (5.9)$$

an integer divisible by q by the chosen congruence. The corrected deficit therefore has the form z/D with $z \in \mathbb{Z}$, so its reduced denominator is coprime to p : the full current p -power has been eliminated. Every newly introduced prime power is smaller than q : those from m because $m < q$, and those from $qm + \sigma_m$ by (3.5). Strict descent below q is all that is needed; no bound $q/2$ is claimed for correction neighbours. If the current denominator has no prime-power factor numerically equal to q , set $S = \emptyset$ and $c_q = 0$.

In either case, subtract next the masses of any $s(q) - c_q$ pairs from $\mathcal{F}_q$. Their denominators introduce only prime powers at most $q/2$. They may reintroduce a smaller power of the same prime p , which is handled at its later turn. After the complete stage every prime power in the denominator is strictly smaller than q , which is the invariant for the next stage, including at stages using only auxiliary pairs. Every prime power in $(L, H]$ is visited, including those at which only padding is needed.

Count and endpoints. Each stage adds exactly $s(q)$ pairs, so $|\mathcal{C}| = C_H$, and $C_H \leq \sum_{n \leq H} n^{1/20} \leq H^{21/20}$. A retained correction pair has endpoints in $\{qm - 1, qm, qm + 1\}$ with $m \leq 8q^\varepsilon$, so they are at most $8q^{11/10} + 1 \leq 8H^{11/10} + 1$, and at least $qm - 1 \geq q - 1 \geq L$; an auxiliary pair has endpoints in $[q^{11/10}, 2q^{11/10}]$, which lie in the same range. Correction endpoints lie in $\mathcal{P}_L^*$ and auxiliary endpoints in $\mathcal{F}^*$, so all endpoints lie in $\mathcal{U}$.

Separation. Retained correction pairs are pairwise separated by Lemma 2; auxiliary pairs are pairwise separated and separated from every possible correction pair by Lemma 5.

Mass. For $qm \geq 2$,

$$w_{-1}(qm) = \frac{1}{qm} + \frac{1}{qm - 1} \leq \frac{3}{qm}, \quad w_{+1}(qm) \leq \frac{2}{qm} \leq \frac{3}{qm}.$$

With $\varepsilon = 1/10$, $m \geq R(q) = q^{1/10}/E(q)$ and $E(q) \leq q^{1/40}$ (condition (f) of Section 2.4),

$$w_{\sigma_m}(qm) \leq 3E(q)q^{-11/10}.$$

Each auxiliary pair has mass at most $2q^{-11/10} \leq 3E(q)q^{-11/10}$, since $E(q) \geq 1$. A stage of exactly $s(q)$ pairs therefore costs at most

$$3s(q)E(q)q^{-11/10} \leq 3q^{1/20+1/40-11/10} = 3q^{-41/40}. \quad (5.10)$$

Summing over all integers $q > L$,

$$W_C \leq 3 \sum_{n > L} n^{-41/40} \leq 3 \int_L^\infty x^{-41/40} dx = 120L^{-1/40} < \frac{\delta}{16} \quad (5.11)$$

by condition (h) of Section 2.4.

Residual. On completion the reduced denominator has only prime powers at most L , each of which divides $K = \text{lcm}(1, \dots, L)$; hence the reduced denominator of r divides K , and $r \geq r_0 - W_C > r_0 - \delta/16 \geq \delta/4 - \delta/16 > 0$. $\square$

6 The core

6.1 Dense bands contain a unit-fraction sum

Lemma 6 (dense bands). *If $T \rightarrow \infty$ through positive integers and $E_T \subseteq [T, 4T) \cap \mathbb{Z}$ has size $o(T)$, then $[T, 4T) \setminus E_T$ has a subset whose reciprocals sum to 1.*

Proof. The density of $[T, 4T) \setminus E_T$ in $[1, 4T]$ tends to $3/4 > 1 - 1/e + 1/20$. Apply Theorem 3 with $\zeta = 1/20$ and $N = 4T$. $\square$

6.2 Core pairs and their extensions

For $k \rightarrow \infty$ put, as in Section 2.4,

$$Q = \lfloor k^{4/5} \rfloor, \quad H = KBQ + 2,$$

and define the set of *reserved parameters*

$$\mathcal{R} = \{d \in [Q, BQ) \cap \mathbb{Z} : [Kd - 1, Kd + 3] \cap \mathcal{U} = \emptyset\}, \quad R = |\mathcal{R}|. \quad (6.1)$$

The protected window $[Kd - 1, Kd + 3]$ contains the one-integer neighbourhood of the possible core triple $[Kd, Kd + 2]$. Thus neither that triple nor its shorter pair $[Kd + 1, Kd + 2]$ can overlap or be adjacent to any pair with both endpoints in $\mathcal{U}$, in particular to any pair in the set $\mathcal{C}$ of Proposition 4, whichever orientations and subsets that proposition uses. The core is protected against the deterministic superset $\mathcal{P}_L^*$ of every possible signed endpoint, not only against a previously selected family.

Proposition 5 (core). *For sufficiently large k the following hold.*

- (i) $R = (B - 1)Q - o(Q)$; in particular $R = o(k)$, and $C_H = O(k^{21/25}) = o(k)$.
- (ii) The core pairs $[Kd + 1, Kd + 2]$, $d \in \mathcal{R}$, are pairwise separated, as are the triples $[Kd, Kd + 2]$, $d \in \mathcal{R}$, and every endpoint is at most H . Their mass satisfies

$$W_{\text{core}} \leq \frac{2}{K} \sum_{Q \leq d < BQ} \frac{1}{d} = \frac{2j_0 \log 4}{K} + o(1) \leq 4\delta \log 4 + o(1) < \frac{1}{8}. \quad (6.2)$$

- (iii) There are pairwise disjoint sets $D_0, \dots, D_{j_0-1} \subseteq \mathcal{R}$ with $\sum_{d \in D_i} 1/d = 1$ for each i . Consequently, for every integer $1 \leq j \leq j_0$, replacing the core pair $[Kd + 1, Kd + 2]$ by the triple $[Kd, Kd + 2]$ for every $d \in D_0 \cup \dots \cup D_{j-1}$ adds exactly j/K to the mass and changes neither the number R of core intervals nor their separation.

Proof. (i) Every excluded $d \in [Q, BQ)$ has a witness $n \in \mathcal{U} \cap [Kd - 1, Kd + 3]$, and these windows are disjoint for distinct d because $K \geq 8$; all witnesses lie in $[1, KBQ + 3]$. Hence the number of excluded parameters is at most $|\mathcal{U} \cap [1, KBQ + 3]| = o(KBQ) = o(Q)$ by (5.6), since K and B are fixed. So $R = (B - 1)Q - o(Q)$, which is $\Theta(k^{4/5}) = o(k)$. By (5.8), $C_H \leq H^{21/20} = O(Q^{21/20}) = O(k^{21/25})$.

(ii) Distinct triples $[Kd, Kd + 2]$ have starts differing by at least $K \geq 8$, so they are separated, and so are the pairs they contain. All endpoints are at most $KBQ + 2 = H$. The mass bound (6.2) uses $\sum_{Q \leq d < BQ} 1/d = \log B + o(1) = j_0 \log 4 + o(1)$ and $j_0/K \leq \delta + 1/K \leq 2\delta$ by condition (i) of Section 2.4.

- (iii) Apply Lemma 6 in the j_0 disjoint bands

$$[4^i Q, 4^{i+1} Q), \quad 0 \leq i < j_0,$$

all of which lie inside $[Q, BQ)$ because $B = 4^{j_0}$. Each band loses only the $o(Q) = o(4^i Q)$ parameters excluded from $\mathcal{R}$, and j_0 is fixed, so Lemma 6 with $T = 4^i Q$ gives $D_i \subseteq \mathcal{R} \cap [4^i Q, 4^{i+1} Q)$ with $\sum_{d \in D_i} 1/d = 1$. The sets D_i are disjoint because the bands are. Extending $[Kd + 1, Kd + 2]$ to $[Kd, Kd + 2]$ adds $1/(Kd)$ to the mass, so extending over $D_0 \cup \dots \cup D_{j-1}$ adds exactly j/K . The extended family is still separated by (ii), and its cardinality is still R . $\square$

7 Main pairs and the exact interval count

By Lemma 4 with $y = Q = k^{4/5+o(1)}$ and $\theta = 4/5$, in any fixed interval $[ak, bk]$ at most

$$((b-a)\log(5/4) + o(1))k$$

integers are not Q -powersmooth. Consider the *grid pairs* $[a, a+1]$ with $3 \mid a$. Distinct grid pairs are separated, and an integer that is not Q -powersmooth spoils at most one of them. Thus $[ak, bk]$ contains at least

$$\gamma(b-a)k + o(k) \text{ grid pairs with both endpoints } Q\text{-powersmooth,} \quad \gamma = \frac{1}{3} - \log \frac{5}{4} > 0, \quad (7.1)$$

where only $O(1)$ pairs are affected by the boundary. Numerically $\gamma = 0.110\dots$, so

$$10\gamma > 1, \quad 16\gamma > 1. \quad (7.2)$$

Proposition 6 (main pairs). *Let $M = k - R - C_H$ and $\tau_k = 1 - W_{\text{core}} - \delta/2$. For sufficiently large k there is a set $\mathcal{M}$ of exactly M grid pairs with both endpoints Q -powersmooth, contained in $[k/1024, k] \cup [4k, 20k]$, pairwise separated, whose mass W_{main} satisfies*

$$\tau_k - 2048/k \leq W_{\text{main}} \leq \tau_k. \quad (7.3)$$

Consequently the initial deficit

$$r_0 = 1 - W_{\text{core}} - W_{\text{main}} \in [\delta/2, \delta/2 + 2048/k] \quad (7.4)$$

is a rational number whose reduced denominator has no prime-power divisor exceeding H .

Proof. By Proposition 5(i), $M = k - o(k)$ is a positive integer for large k . Let $\mathcal{A}$ be the set of all grid pairs with both endpoints Q -powersmooth contained in $[k/1024, k]$. Divide this interval into the ten dyadic bands $[t, 2t]$, $t = 2^{-i}k$, $1 \leq i \leq 10$. By (7.1) each band contributes at least $\gamma t + o(k)$ pairs, each of mass at least $1/t$, so

$$\sum_{A \in \mathcal{A}} \sum_{n \in A} \frac{1}{n} \geq 10\gamma + o(1) > 1 \quad (7.5)$$

by (7.2). There are fewer than $k/3$ pairs in $\mathcal{A}$, hence fewer than M .

The far band $[4k, 20k]$ contains more than k grid pairs with both endpoints Q -powersmooth, by (7.1) and $16\gamma > 1$. Choose any M of them. Their total mass is at most $M/(2k) < 1/2$.

Now replace far pairs, one at a time, by the pairs of $\mathcal{A}$, until every pair of $\mathcal{A}$ has been inserted. Each replacement removes a pair of mass less than $1/(2k)$ and inserts one of mass at most $2048/k$ and at least $2/(k+1)$, so it increases the mass by at most $2048/k$. By (7.5) the mass at the end exceeds 1. Since $W_{\text{core}} < 1/8$ by (6.2), the target τ_k lies in $(1/2, 1)$ for large k . Stop at the last configuration before the mass first exceeds τ_k ; this gives (7.3). The count is exactly M throughout. The pairs are separated within each band by the grid, and the two bands are separated from one another.

Finally, (7.4) follows from (7.3). Every main endpoint is Q -powersmooth, and every core endpoint is at most H by Proposition 5(ii), so all prime powers in the reduced denominator of r_0 are at most H . $\square$

8 Proof of the main theorem

Let k be sufficiently large for Propositions 5 and 6 and for the estimates below, and put $Q = \lfloor k^{4/5} \rfloor$, $H = KBQ + 2$ and C_H as in (5.8). Construct, in this order:

1. the core pairs $[Kd + 1, Kd + 2]$, $d \in \mathcal{R}$, and the sets $D_0, \dots, D_{j_0-1}$ of Proposition 5; these depend only on $\mathcal{U}$, not on any orientation choice;
2. the compatible fibers $(I_q, \sigma_q)_{L < q \leq H}$ of Lemma 2 for this H , which is finite; condition (b) of Section 2.4 gives $L \geq L_0$;
3. the $M = k - R - C_H$ main pairs $\mathcal{M}$ of Proposition 6, with the initial deficit r_0 of (7.4);
4. the C_H correction and auxiliary pairs $\mathcal{C}$ of Proposition 4, applied to r_0 , H and the fibers of step 2. The hypotheses hold: $H > L$ for large k , $r_0 \geq \delta/2 \geq \delta/4$, and the denominator of r_0 has prime powers at most H by Proposition 6.

The residual $r = r_0 - W_{\mathcal{C}}$ satisfies, by Proposition 4(iii)–(iv) and (7.4),

$$\frac{\delta}{4} < \frac{\delta}{2} - \frac{\delta}{16} \leq r \leq \frac{\delta}{2} + \frac{2048}{k} \leq \delta$$

for large k , and its reduced denominator divides K . Hence $r = j/K$ with an integer $1 \leq j \leq K\delta \leq j_0$. Extend the core pairs indexed by $D_0 \cup \dots \cup D_{j_0-1}$ to triples, as in Proposition 5(iii). The total mass is now

$$W_{\text{core}} + W_{\text{main}} + W_{\mathcal{C}} + \frac{j}{K} = (1 - r_0) + W_{\mathcal{C}} + r = 1.$$

Count. The family consists of R core intervals, M main pairs and C_H pairs from $\mathcal{C}$; no interval was split or merged, so the total number is $R + M + C_H = k$. Every interval has two or three members.

Separation. The pairs of $\mathcal{C}$ are pairwise separated by Proposition 4(i). Their endpoints lie in $\mathcal{U}$, so by the definition (6.1) of $\mathcal{R}$ every core interval, extended or not, is separated from every pair of $\mathcal{C}$, for both orientations and both signs of the correction pairs. Distinct core intervals are separated by Proposition 5(ii). Main pairs are separated from one another by Proposition 6. Finally, all endpoints of $\mathcal{C}$ are at most

$$8H^{11/10} + 1 = O(k^{22/25}) = o(k),$$

and all core endpoints are at most $H = O(k^{4/5}) = o(k)$, so for large k all of them lie below $k/1024 - 2$, separating them from every main pair. Thus the whole family is separated.

Endpoints. Main endpoints are at most $20k$ and all other endpoints are $o(k)$, so $b_i \leq 20k$. Every left endpoint is at least $\min\{L, KQ, k/1024\} \geq 2$: the endpoints of $\mathcal{C}$ are at least L by Proposition 4(ii), core endpoints are at least KQ , and main endpoints are at least $k/1024$.

Listing the intervals in increasing order gives exactly the conditions (1.1). This proves Theorem 1. $\square$

Acknowledgements and declaration

Large language models were used extensively in this work: in developing and checking the argument, in drafting this manuscript, and in writing the accompanying Lean formalization. The models

used were GPT 6 Astra (OpenAI), Claude Fable 5.1 (Anthropic) and Gemini 3.8 Flash (Google). The Lean formalization builds on Mathlib and, in addition, on Boris Alexeev’s `plby/lean-proofs` library of machine-generated Lean proofs [Al], roughly three million lines, from which the modules supplying the structure theorem, the Liu–Sawhney theorem, Mertens’ theorem and the divisor bound are vendored with their provenance recorded (Appendix A). All mathematical content was reviewed by the author, who takes sole responsibility for it. The author thanks the reviewers whose comments on earlier drafts led to the present formulation.

A Formal verification certificate

A.1 The formal target

The audited file `Erdos289/Intervals.lean` declares, inside the namespace `Erdos289`, the following two items. The display is a schematic ASCII rendering of the declaration: the source uses Unicode ($\mathbb{N}$, $\rightarrow$, $\forall$, $\exists$, $\leq$, $\vee$, $\sum$), and the exact text is in the file and is reproduced in the repository README.

```
structure FamilyWitness (k : Nat) where
  intervals : Fin k -> NatInterval
  positive : forall i, 1 <= (intervals i).lo
  short : forall i, (intervals i).hi = (intervals i).lo + 1 \ /
    (intervals i).hi = (intervals i).lo + 2
  separated : forall i j, i < j -> (intervals i).Separated (intervals j)
  bounded : forall i, (intervals i).hi <= 20 * k
  total_mass : sum i, (intervals i).mass = 1

def CandidateStatement : Prop :=
  exists k0 : Nat, forall k : Nat, k0 <= k -> Nonempty (FamilyWitness k)
```

Here `NatInterval` is a pair of natural-number endpoints `lo`, `hi`; its carrier is the set of integers from `lo` to `hi` inclusive; `I.Separated J` means `I.hi + 1 < J.lo`; and `I.mass` is the exact rational sum of the reciprocals of the carrier. The field `positive` excludes the denominator zero; `short` gives exactly two or three members; `separated` with indices in `Fin k` gives exactly k pairwise separated intervals; `total_mass` is an exact equality in $\mathbb{Q}$. This is Theorem 1 with the lower endpoint condition $a_i \geq 1$ in place of $a_i \geq 2$; Remark 1 shows the two are equivalent given the other conditions. The intervals are indexed by $\{0, \dots, k-1\}$ and are separated for every ordered pair of indices, which after sorting is the condition $b_i + 1 < a_{i+1}$.

A.2 Certificate metadata

Repository	https://github.com/beetree/math_erdos_289
Commit	61f54d7ba430b3869854bb51ccebfb16e48e5c11d (branch main)
Earlier certificate	tag v1.0-unconditional-certificate, commit 6de963b7572b9ea27aa3339de85dc7254744687f, an earlier certificate for the same terminal target with the same target file, archival file and manifest hashes. Between the two revisions the Lean sources changed in the compatibility file <code>Erdos289/Compat.lean</code> (which the terminal theorem does not import), in comments, and by the renaming of one vendored file to an ASCII name; the git history records the exact diff
Lean toolchain	leanprover/lean4:v4.34.0-rc2 (the file <code>lean-toolchain</code>)
Mathlib	v4.34.0-rc2, resolved revision 85e3a25e006c35636f0e53b0e9296caca2685bc0
Other Lake dependencies	as resolved in <code>lake-manifest.json</code> at the tag: batteries d54dddc5, aesop 18889deb, Qq 507746ab, proofwidgets a8acbfd8, importGraph d8823026, LeanSearchClient ba67e212, plausible d9598f07, Cli ab3a82db. The manifest at the tag has SHA-256 119e31567cce06a9f16a6fe8dd2fab7636ce8483c1a2687efc38ced5bc53e773.
Vendored proofs	modules under <code>SolveMath/</code> (Liu–Sawhney, Mertens, divisor bound; 41 modules) and <code>ErdosProblems/</code> (the structure theorem, from the Erdős 186 development; 258 modules), ported from the recorded reference snapshot of <code>plby/lean-proofs</code> [Al], commit 61fce10ef6671b1df0325f22bc76c8cd1f2fa554, with the listed compatibility and warning-cleanup edits. Provenance, upstream authorship and every deviation are recorded in <code>ErdosProblems/PROVENANCE.md</code> and <code>SolveMath/PROVENANCE.md</code> . Of the 258 ported structure-theorem files, 255 carry Apache-2.0 headers; the three exceptions and the upstream license material are documented in the provenance record.
Bridges	the structure input is supplied by <code>Erdos289.Ported.cfhmprsv_structure_audited</code> in <code>Erdos289/CFPBridge.lean</code> . The other audited statements are supplied by <code>Erdos289.Ported.liu_sawhney_audited</code> , <code>Erdos289.Ported.mertens_second_audited</code> and <code>Erdos289.Ported.divisor_bound_audited</code> in <code>SolveMath/Ported/</code> . The working aliases used by the construction (<code>Erdos289.liu_sawhney</code> , <code>mertens_second</code> , <code>divisor_bound</code> , <code>cfhmprsv_structure</code>) are connected to these proved inputs in <code>Erdos289/ExternalBridge.lean</code> . Earlier axiom-based bridge declarations remain in that file but are not the active aliases.
Terminal declaration	<code>Erdos289.candidateStatement</code> (file <code>Erdos289/Main.lean</code>)
Terminal type	<code>Erdos289.CandidateStatement</code>
Target file	<code>Erdos289/Intervals.lean</code> , SHA-256 178e26470eb61a81f183761d053b697d1800bda64a622dd69858ad065f441871, unchanged since the statement audit
Archival specification	<code>Erdos289/ExternalAxioms.lean</code> , SHA-256 6a4affa2cecb731c87ec21433305842bd3c0c92bc7c62a8ca86f410f3647c5d7; its six assumed declarations are not in the dependency chain
Axioms	[<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>]
Build	warning-free lake build, with the axiom report printed by <code>Erdos289/Main.lean</code> ; transcript below

The following transcript (long lines wrapped for the page) was recorded on 4 September 2026 in the maintainer’s working copy at the commit above, with a clean status, the pinned toolchain and the existing lockfile; `lake build` replays the up-to-date modules from the build cache.

```
$ git rev-parse HEAD
61f54d7ba430b3869854bb51ccebfb16e48e5c11d
$ cat lean-toolchain
leanprover/lean4:v4.34.0-rc2
$ sha256sum Erdos289/Intervals.lean lake-manifest.json
178e26470eb61a81f183761d053b697d1800bda64a622dd69858ad065f441871
  Erdos289/Intervals.lean
119e31567cce06a9f16a6fe8dd2fab7636ce8483c1a2687efc38ced5bc53e773
```



```

lake-manifest.json
$ lake build
i [9101/9104] Replayed Erdos289.Main
info: Erdos289/Main.lean:25:0: 'Erdos289.candidateStatement' depends on
  axioms: [propext, Classical.choice, Quot.sound]
Build completed successfully (9104 jobs).
$ lake env lean AxiomCheck.lean
Erdos289.candidateStatement : Erdos289.CandidateStatement
'Erdos289.candidateStatement' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'Erdos289.Ported.cfhmppsv_structure_audited' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'Erdos289.Ported.liu_sawhney_audited' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'Erdos289.Ported.mertens_second_audited' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'Erdos289.Ported.divisor_bound_audited' depends on axioms:
  [propext, Classical.choice, Quot.sound]

```

Here AxiomCheck.lean is the file

```

import Erdos289.Main
#check (Erdos289.candidateStatement : Erdos289.CandidateStatement)
#print axioms Erdos289.candidateStatement
#print axioms Erdos289.Ported.cfhmppsv_structure_audited
#print axioms Erdos289.Ported.liu_sawhney_audited
#print axioms Erdos289.Ported.mertens_second_audited
#print axioms Erdos289.Ported.divisor_bound_audited

```

whose output is shown; the last four reports are diagnostics for the retained theorem interfaces, and the terminal report is the essential gate. The build produced no errors and no warnings. Earlier the same day a from-scratch build in a fresh clone of the tagged revision (`lake exe cache get` for Mathlib, every project module compiled from source) produced the same reports for that revision.

Audit scope. The transcript above was produced on the author’s side, on the maintainer’s machine; it is not a reproduction by an independent third party, and no such reproduction has yet been reported. A reviewer who audited the source tree of this revision without a Lean toolchain confirmed the hashes and the import structure but could not run the build. To reproduce, start from a fresh checkout of the commit above with its existing lockfile, do not update dependency revisions, run `lake exe cache get` and then the commands above. The exact target type and the complete axiom report of the lowercase theorem `Erdos289.candidateStatement` are the relevant mathematical checks; printing the axioms of the proposition `Erdos289.CandidateStatement` is not the proof audit, and a warning-free build is hygiene. An unused axiom declaration elsewhere in the repository does not make the terminal theorem conditional.

A.3 Correspondence between the paper and the formalization

The kernel establishes the formal implications. The remaining human task is to check that the formal proposition says what the problem and this paper claim (Appendix A.1), and, for readers

comparing the two proofs, that the formal components correspond to the components here. The following points were checked for that correspondence.

Component	Correspondence
Signed pairs	the sign is exactly ± 1 ; the interval is the ordered pair $\min / \max$ of qm and $qm + \sigma$; both denominators are positive (signedPair , wSigned).
Multiplier range	$R(q) \leq m \leq 8q^\varepsilon$ with V, E, R as in (3.1) (Venv , Eenv , Rq); the envelope thresholds are eventual statements (venv_le , Eenv_le_eventually).
Strict powersmoothness	every prime power dividing the neighbour is $< q$, including powers of small primes (Powersmooth ($q - 1$)).
Compatible fibers	quantifier order $\forall \varepsilon \exists L_0 \forall L \geq L_0 \forall H \geq L \exists (I_q, \sigma_q)$ (lemmaF2); sets of distinct multipliers; all cross-label intervals separated (Iv.Sep).
Position orientation	the bit chooses left or right relative to the multiple of four (slot), not a uniform sign.
Inverse covering	$C = 8$ (lemma3_wide); units modulo the full q ; a subset, not a multiset; at most $\lfloor q^{\varepsilon/2} \rfloor$ summands.
Structure-theorem bridge	the same represented progression; nonempty proper boxes at scales 1 and cs (properAt); source-native $\log_2$ (logTwo); threshold $m \geq m_0$ and the cardinality quantifiers.
Descent	all stages through H ; exact padding (CH); the signed common-denominator identity (5.9); the whole current p -power removed before smaller powers are handled (descentS).
Core protection	avoids the deterministic superset $\mathcal{P}_L^*$ of every possible signed endpoint (PstarSigned , USigned), not only a selected family (coreS).
Final count and mass	exactly $R + M + C_H = k$; core extensions preserve the count; total reciprocal mass exactly one (erdos289S2).

The paper and the formalization prove the same covering lemma with different intermediate estimates. The formalization uses an explicit fixed power loss in the divisor-count step: the lemma **paper_steps_4_5** (file **Erdos289/Lemma3Steps45.lean**) proves, from the retained-set estimate $|J| \geq q^{7\varepsilon/8}/2$ and a divisor majorant of the form $\max\{n_0, x^{\varepsilon/4000}\}$,

$$V_{\text{GAP}} \geq \frac{q^{1-d\varepsilon/8-d\varepsilon/500}}{(16Cd)^d}.$$

This differs from the divisor-envelope bound (4.5), which uses $|J| \geq 3m/4$ and $\Delta_C(q)$, and neither bound is claimed to be the other; but it is sufficient for the same rank conclusion. For $d \geq 2$ its exponent exceeds the upper-bound exponent of (4.3) by

$$\varepsilon \left(\frac{3d-4}{8} - \frac{d}{500} \right) > 0,$$

which at $d = 2$ is $\varepsilon(1/4 - 1/250)$ and increases with d ; for $d = 1$ the progression-length exponent is $1 + (3/8 - 1/500)\varepsilon > 1$. Both positivity claims are uniform in the bounded rank $d \leq d_0$. The signed descent uses the common mass bound (5.10) summed as in (5.11), giving $120L^{-1/40}$ in both presentations. Other harmless differences: the formal odd-prime construction inverts $4t$ rather than solving $4rt \equiv -1$, which exchanges the two candidates; the formal first sieve uses $\eta = \varepsilon/20000$ with allowance $M_q/200$; the orientation count averages $(81/256)^{B_q}$ rather than 2^{-B_q} ; the density proof splits the labels at a smaller exponent; the cancellation uses a product denominator rather than a least common multiple; and the auxiliary avoidance condition is stronger than the displayed distance two. None of these changes the terminal target or adds an axiom, and the proof scripts are not a line-for-line translation of every displayed estimate.

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